Choice-Free Reverse Mathematics.

From Barrow's Geometry to the Recursive Comprehension Axiom RCA_0

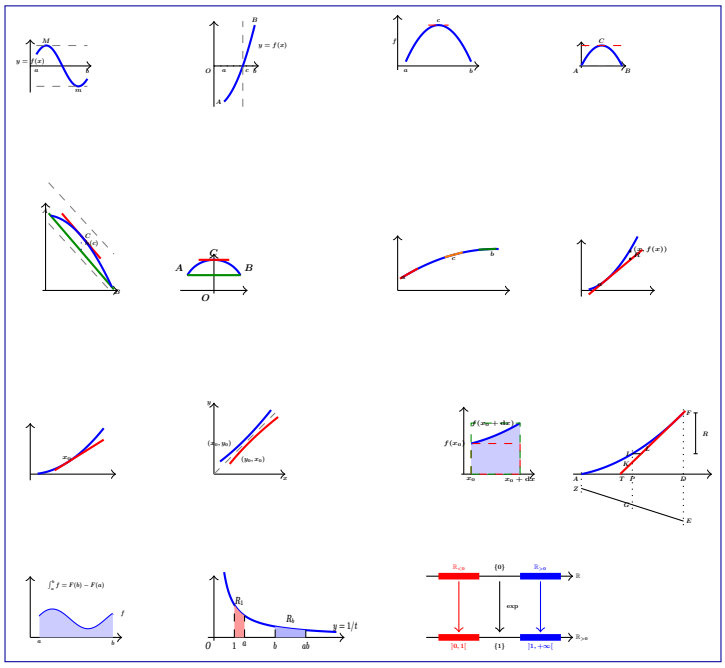
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Abstract

Part I develops classical single-variable real analysis in Barrow's geometric language of areas, ordinates, and subtangents. Part II classifies each theorem in reverse mathematics. MVT and Rolle are RCA_0 via Hardin–Velleman; the mean-value theorem for integrals, the inverse-function theorem (Yokoyama's strong-C^1 form), and the ln/exp chain are RCA_0; IVT, EVT, Darboux, and the weighted MVT are WKL_0 as proved here. All C^k hypotheses are read as strong C^k-coding. No entry uses Cauchy completeness or any form of Choice.



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Introduction

Barrow's *Geometrical Lectures* (1670) give the first calculus textbook and the first geometric proof of the Fundamental Theorem of Calculus; see [1, 2].

Part I develops the classical theorems of single-variable real analysis — EVT, IVT, Fermat, Rolle, MVT and its refinements, convexity, inverse function, the Fundamental Theorems of Calculus, and the ln/exp addition and image theorems — in Barrow's geometric idiom, with figures and geometric proofs.

Part II classifies each theorem in reverse mathematics. Every axiom, postulate, definition and statement of Part I reappears as a tagged entry with a dependency table and an exact strength. Classification: all entries are RCA_0 except IVT, EVT, Darboux, and the weighted MVT, which are WKL_0. No entry uses Cauchy completeness or any form of Choice [7, 14]. Coding conventions are in §5.

Part 1. A Geometric Development of Classical Real Analysis

Following Barrow [1, Lecture X, §11], we state and prove the classical theorems in Barrow's geometric idiom. See [2] for commentary.

1. Axioms, definitions and geometric conventions

Let $[a,\,b] \subseteq \mathbb{R}$ and let $f\colon [a,\,b] \to \mathbb{R}$ be continuous. The *area function* is

$$A_a^x := \text{(signed) area of } \{(t,\,y) : a \le t \le x,\ \ 0 \le y \le f(t)\}, \tag{1}$$

with Barrow's orientation

$$A_x^x = 0, \qquad A_x^{x+\mathrm{d}x} = -A_{x+\mathrm{d}x}^{x} \quad (\mathrm{d}x < 0). \tag{2}$$

The curve $u \mapsto A_a^u$ is Barrow's AIF; $f(x_0)$ is his DE; the unit is his R.

Differentiability. f is differentiable at x if

$$\lim_{h\to 0} (f(x+h) - f(x))/h = f'(x) \tag{3}$$

exists.

Tangency (little-*o* form). G is differentiable at x_0 with tangent $L(u) = m(u - x_0) + G(x_0)$ iff

$$G(x_0 + \Delta x) = G(x_0) + m\,\Delta x + \Delta x \cdot o(\Delta x) \quad (\Delta x \to 0), \tag{4}$$

in which case $G'(x_0) = m$; cf. [2, Theorem 2].

Lin-Integ.

$$\forall u,\, v,\, w \in [a,\, b] :\ A_u^w = A_u^v + A_v^w. \tag{5}$$

Int-Mono.

$$f \le g \text{ on } [a,\, b] \Rightarrow \int_a^b f \le \int_a^b g. \tag{6}$$

Similarity. For $b > 0$,

$$\int_b^{ab} h(t)\,\mathrm{d}t = \int_1^a h(bu)\,b\,\mathrm{d}u. \tag{7}$$

Inv-Cont.

$$t \mapsto 1/t \in C^0(\mathbb{R}_{>0}). \tag{8}$$

Image-Union.

$$f(A \cup B) = f(A) \cup f(B). \tag{9}$$

Product rule.

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x). \tag{10}$$

Chain rule.

$$(f \circ g)'(x) = f'(g(x))\,g'(x). \tag{11}$$

Sign-Law.

$$\forall u > 0:\ (\ln u < 0 \iff u < 1) \wedge (\ln u = 0 \iff u = 1) \wedge (\ln u > 0 \iff u > 1). \tag{12}$$

Ln-Bij. $\ln : \mathbb{R}_{>0} \to \mathbb{R}$ is bijective; $\exp := \ln^{-1} : \mathbb{R} \to \mathbb{R}_{>0}$, i.e.

$$\exp := \ln^{-1} : \mathbb{R} \to \mathbb{R}_{>0}. \tag{13}$$

Barrow-tangency. Let G be differentiable at x_0, and let $L(u) = m(u - x_0) + G(x_0)$ be the line through $(x_0,\ G(x_0))$ of slope m. If $L \le G$ (or $L \ge G$) on a punctured neighbourhood of x_0, then

$$m = G'(x_0). \tag{14}$$

[1]

2. Statements of the classical theorems

2.1. Topological theorems

Theorem 1 (Extreme Value Theorem).
If $f : [a,\ b] \to \mathbb{R}$ is continuous, then

$$\exists x_{\max},\ x_{\min} \in [a,\ b],\ \ \forall x \in [a,\ b]:\ \ f(x_{\min}) \le f(x) \le f(x_{\max}). \tag{15}$$

Theorem 2 (Intermediate Value Theorem).
If $f : [a,\ b] \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$, then

$$\exists c \in (a,\ b):\ \ f(c) = 0. \tag{16}$$

[1] The two-sided hypothesis strengthens Barrow's one-sided criterion, which fails at inflections [2]. Every use in Part I is in a convex or concave setting, where the two-sided hypothesis holds.

2.2. Differentiation theorems

Theorem 3 (Fermat).

If f has a local extremum at $c \in (a,\, b)$ and is differentiable at c, then

$$f'(c) = 0. \tag{17}$$

Theorem 4 (Rolle).

If f is continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $f(a) = f(b)$, then $\exists c \in (a,\, b) :\ f'(c) = 0$.

Theorem 5 (Mean-Value Theorem).

If f is continuous on $[a,\, b]$ and differentiable on $(a,\, b)$, then

$$\exists c \in (a,\, b) :\ f'(c) = \frac{f(b) - f(a)}{b - a}. \tag{18}$$

Geometrically, $d_c \parallel d_{a,\, b}$.

Theorem 6 (Constant-function).

If f is continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $f' = 0$ on $(a,\, b)$, then

$$f(x) = f(a) \quad \text{for all } x \in [a,\, b]. \tag{19}$$

Theorem 7 (Monotonicity).

If f is continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $f' \geq 0$ (resp. $f' > 0$), then

$$x < y \Rightarrow f(x) \leq f(y) \quad (\text{resp. } f(x) < f(y)) \tag{20}$$

for all $x,\, y \in [a,\, b]$.

Theorem 8 (Cauchy MVT).

If $f,\, g$ are continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $g' \neq 0$ on $(a,\, b)$, then

$$\exists c \in (a,\, b) :\ \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}. \tag{21}$$

Theorem 9 (Darboux).

If f is differentiable on $[a,\, b]$ and $f'(a) < \lambda < f'(b)$, then

$$\exists c \in (a,\, b) :\ f'(c) = \lambda. \tag{22}$$

Theorem 10 (L'Hôpital).

Let $f,\, g$ be differentiable near a, with $f(a) = g(a) = 0, g' \neq 0$ near a, and $\lim_{x \to a} f'(x)/g'(x) = L$. Then

$$\lim_{x \to a} f(x)/g(x) = L. \tag{23}$$

Theorem 11 (Taylor, Lagrange remainder).
If f is $(n+1)$-times differentiable on an interval containing a and x, then

$$\exists c \text{ between } a,\ x:\ \ f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}. \tag{24}$$

Theorem 12 (Convexity).
If f is twice differentiable on $[a,\ b]$ and $f'' \geq 0$ on $(a,\ b)$, then

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0) \quad \text{for all } x,\ x_0 \in [a,\ b]. \tag{25}$$

Theorem 13 (Inverse function).
If $f : I \to \mathbb{R}$ is continuous and strictly monotone on I, x_0 lies in the interior I° of I, and f is differentiable at x_0 with $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ with

$$(f^{-1})'(y_0) = 1/f'(x_0). \tag{26}$$

2.3. Integration theorems

Lemma 14 (Barrow's area inequality).
If f is continuous and increasing on $[a,\ b]$ and $x_0,\ x_0 + \mathrm{d}x \in [a,\ b], \mathrm{d}x \neq 0$, then

$$f(x_0)\,\mathrm{d}x \leq A_{x_0}^{x_0+\mathrm{d}x} \leq f(x_0 + \mathrm{d}x)\,\mathrm{d}x, \tag{27}$$

and both inequalities are strict when f is strictly increasing on the interval with endpoints x_0 and $x_0 + \mathrm{d}x$.

Lemma 15 (MVT for integrals).
If f is continuous on $[a,\ b]$ and $x_0,\ x_0 + \mathrm{d}x \in [a,\ b], \mathrm{d}x \neq 0$, then

$$\exists \theta \text{ between } x_0,\ x_0 + \mathrm{d}x:\ \ A_{x_0}^{x_0+\mathrm{d}x} = f(\theta)\,\mathrm{d}x. \tag{28}$$

Theorem 16 (First FTC; Barrow).
If f is continuous on $[a,\ b]$, then

$$\frac{d}{dx}\int_a^x f = f(x) \tag{29}$$

for every $x \in (a,\ b)$, with one-sided versions at $a,\ b$.

Theorem 17 (Second FTC).
If F is differentiable on $[a,\ b]$ and $F' = f$ is continuous, then

$$\int_a^b f = F(b) - F(a). \tag{30}$$

Theorem 18 (Integration by parts).
If $f,\ g \in C^1([a,\ b])$, then

$$\int_a^b fg' = [fg]_a^b - \int_a^b f'g. \tag{31}$$

Theorem 19 (Change of variables).
If $g \in C^1([a,\,b])$ and f is continuous on $g([a,\,b])$, then

$$\int_{g(a)}^{g(b)} f = \int_a^b f(g)\,g'. \tag{32}$$

Theorem 20 (Weighted MVT).
If f is continuous on $[a,\,b]$, $w \geq 0$ is Riemann integrable, and $\int_a^b w > 0$, then

$$\exists c \in [a,\,b] : \int_a^b fw = f(c)\int_a^b w. \tag{33}$$

2.4. The logarithm and the exponential

Theorem 21 (Log-Property).
With $\ln(x) := \int_1^x \mathrm{d}t/t$ $(x > 0)$,

$$\forall a,\,b > 0 :\ \ln(a) + \ln(b) = \ln(ab). \tag{34}$$

Theorem 22 (Exp-Property).
With $\exp := \ln^{-1}$,

$$\forall u,\,v \in \mathbb{R} :\ \exp(u)\exp(v) = \exp(u+v). \tag{35}$$

Theorem 23 (Exp-Image).

$$\exp(\mathbb{R}) = \mathbb{R}_{>0}, \qquad \ln(\mathbb{R}_{>0}) = \mathbb{R}. \tag{36}$$

3. Geometrical proofs

3.1. Topological theorems

Proof of Theorem 1. $S = f([a,\,b])$ is nonempty and bounded; let $M = \sup S, m = \inf S$. Were M unattained, $1/(M - f)$ would be continuous and unbounded on $[a,\,b]$; contradiction. So $M = f(x_{\max})$; symmetrically $m = f(x_{\min})$, giving (15). □

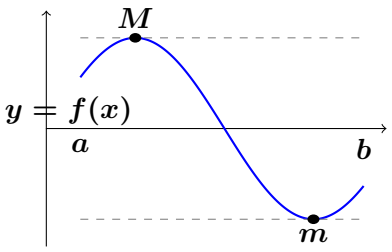


Figure 1. EVT.

Proof of Theorem 2. Assume $f(a) < 0 < f(b)$ and let $c = \sup\{x \in [a,\,b] : f(x) < 0\}$. By continuity $f(c) \leq 0$; if $f(c) < 0$ then $f < 0$ on a neighbourhood of c intersected with $[a,\,b]$, contradicting the definition of c as an upper bound; hence (16) holds. □

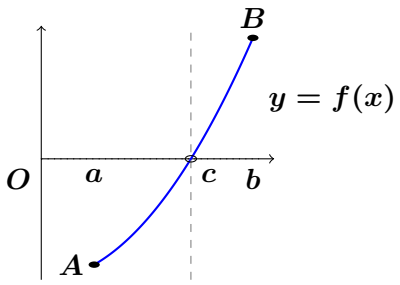


Figure 2. IVT.

3.2. Differentiation theorems

Proof of Theorem 3. If c is a local max: $(f(c+h)-f(c))/h \le 0$ for $h > 0^+$ gives $f'(c) \le 0$; for $h < 0^-, \ge 0$ gives $f'(c) \ge 0$. So $f'(c) = 0$. □

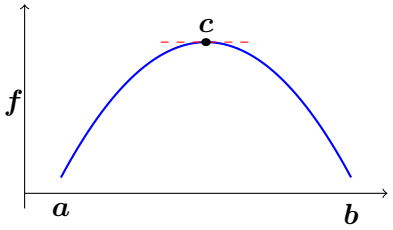


Figure 3. Fermat.

Proof of Theorem 4. By EVT, f attains M, m. If $M = m, f$ constant. Otherwise, $f(a) = f(b)$ forces an interior extremum c; Fermat gives $f'(c) = 0$. □

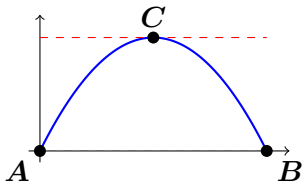


Figure 4. Rolle.

Proof of Theorem 5 (Barrow's sliding tangent). Let L be the chord line, $g = f - L$. Then $g(a) = g(b) = 0$; by EVT, g attains M, m. *Case* $M > 0$*:* maximum at $c \in (a, b)$; the translate $L + M$ touches f at C and lies above; by concave Barrow-tangency, slope $= f'(c) = L'(c) = (f(b)-f(a))/(b-a)$. *Case* $M = 0$*:* if $f = L$ identically, trivial; otherwise $m < 0$ with minimum at interior c and the same argument via convex Barrow-tangency. So $d_c \parallel d_{a,b}$. □

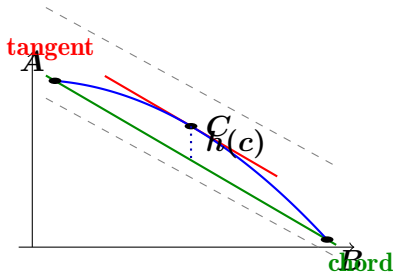


Figure 5. MVT sliding tangent.

Proof of Theorem 6. MVT on $[a, x]$: $f(x) - f(a) = f'(c)(x - a) = 0$. □

Proof of Theorem 7. MVT on $[x_1, x_2]$: $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$; sign of $f'(c)$ gives the result. □

Proof of Theorem 8. $h := (f(b) - f(a))g - (g(b) - g(a))f$ satisfies $h(a) = h(b)$, so Rolle gives c with $(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$; g' and $g(b) - g(a)$ are nonzero, so (21). □

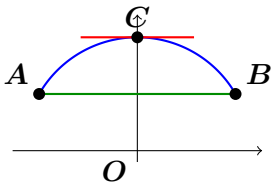


Figure 6. Cauchy MVT.

Proof of Theorem 9. $g = f - \lambda x$ has $g'(a) < 0 < g'(b)$. Small h gives $g(a+h) < g(a)$ and $g(b-h) < g(b)$; by EVT the minimum of g is interior, at c. Fermat: $g'(c) = 0$, i.e. $f'(c) = \lambda$. □

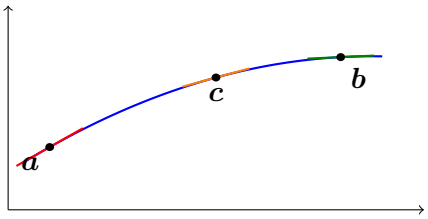


Figure 7. Darboux.

Proof of Theorem 10. Cauchy MVT on $[a, x]$: $f(x)/g(x) = f'(c_x)/g'(c_x)$; $c_x \to a$ as $x \to a^+$. Symmetric for $x \to a^-$. So the limit is L. □

Proof of Theorem 11. $P := \sum f^{(k)}(a)(t-a)^k/k!$; choose M with $f(x) = P(x) + M(x - a)^{n+1}$; $h = f - P - M(t-a)^{n+1}$ has $h^{(j)}(a) = 0, h(x) = 0$. Rolle $n+1$ times: $h^{(n+1)}(c) = 0$, so $M = f^{(n+1)}(c)/(n+1)!$. □

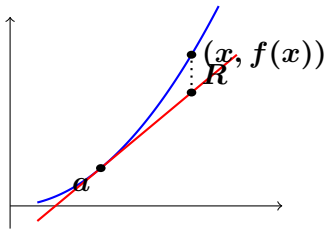


Figure 8. Taylor ($n = 1$).

Proof of Theorem 12. $g = f - f(x_0) - f'(x_0)(x - x_0)$ has $g'' \geq 0$; g' nondecreasing and $g'(x_0) = 0$, so $g' \leq 0$ before and ≥ 0 after x_0; with $g(x_0) = 0, g \geq 0$. Hence (25). □

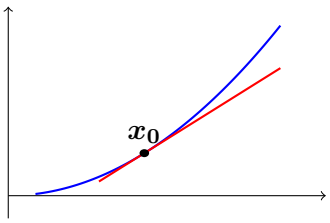


Figure 9. Convexity.

Proof of Theorem 13. Let $c := f'(x_0) \neq 0$. Since $x_0 \in I^\circ$, choose $\rho > 0$ with $(x_0 - \rho, x_0 + \rho) \subseteq I$. *Local surjectivity.* Assume without loss of generality that f is strictly increasing on I (the strictly decreasing case is symmetric). Choose $x_0^-, x_0^+ \in (x_0 - \rho, x_0 + \rho)$ with $x_0^- < x_0 < x_0^+$; then $f(x_0^-) < y_0 < f(x_0^+)$. By the intermediate value theorem applied to the continuous function f on the closed subinterval $[x_0^-, x_0^+] \subseteq I$, every $y \in (f(x_0^-), f(x_0^+))$ is attained at some $x \in (x_0^-, x_0^+) \subseteq I$. Hence $(f(x_0^-), f(x_0^+)) \subseteq f(I)$ is an open neighbourhood of y_0, and f^{-1} is defined on it.

Continuity of f^{-1} at y_0. Differentiability of f at x_0 gives, for some $\delta_0 \in (0, \rho)$ and all x with $|x - x_0| < \delta_0$,

$$|f(x) - f(x_0) - c(x - x_0)| < \tfrac{|c|}{2}|x - x_0|,$$

whence by the reverse triangle inequality $|f(x) - f(x_0)| \geq \tfrac{|c|}{2}|x - x_0|$ on the same neighbourhood. Given $\varepsilon > 0$, set $\varepsilon' := \min(\varepsilon, \delta_0)$ and

$$\eta := \min\Big(\tfrac{|c|}{2}\varepsilon',\ y_0 - f(x_0^-),\ f(x_0^+) - y_0\Big) > 0.$$

Suppose $|y - y_0| < \eta$. Since $y \in (f(x_0^-), f(x_0^+)) \subseteq f(I)$, the value $x := f^{-1}(y) \in (x_0 - \rho, x_0 + \rho) \subseteq I$ is defined. In the case $|x - x_0| < \delta_0$, the bound above gives $|y - y_0| \geq \tfrac{|c|}{2}|x - x_0|$, so $|x - x_0| < \varepsilon' \leq \varepsilon$. In the case $|x - x_0| \geq \delta_0$, let $x_1 := x_0 + \operatorname{sgn}(x - x_0)\,\delta_0 \in (x_0 - \rho, x_0 + \rho) \subseteq I$ be the point of I at distance exactly δ_0 from x_0 on the side of x. By strict monotonicity of f, $|y - y_0| \geq |f(x_1) - f(x_0)|$. Passing the bound $|f(u) - f(x_0)| \geq \tfrac{|c|}{2}|u - x_0|$, valid for $|u - x_0| < \delta_0$, to the limit $u \to x_1$ from inside (x_0, x_1) (using continuity of f at x_1)

yields $|f(x_1)-f(x_0)| \geq \frac{|c|}{2}\delta_0 \geq \frac{|c|}{2}\varepsilon' \geq \eta$, contradicting $|y-y_0|<\eta$. Hence $|y-y_0|<\eta \Rightarrow |f^{-1}(y)-x_0|<\varepsilon$, so f^{-1} is continuous at y_0.

Derivative of f^{-1} at y_0. Changing variable $y=f(x)$, i.e. $x=f^{-1}(y)\to x_0$ as $y\to y_0$ (continuity of f^{-1} at y_0, just established, justifies the substitution),

$$(f^{-1})'(y_0)=\lim_{y\to y_0}\frac{f^{-1}(y)-f^{-1}(y_0)}{y-y_0}=\lim_{x\to x_0}\frac{x-x_0}{f(x)-f(x_0)}=\frac{1}{f'(x_0)},$$

which is (26). □

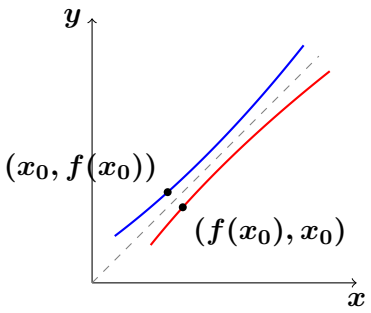


Figure 10. Inverse.

3.3. Integration theorems

Proof of Lemma 14. For $\mathrm{d}x>0$, $f(x_0)\leq f\leq f(x_0+\mathrm{d}x)$ on the interval; inscribed/circumscribed rectangles bound A. For $\mathrm{d}x<0$, use (2). □

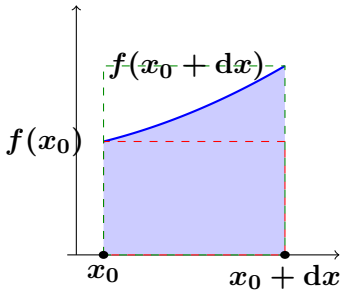


Figure 11. Barrow's area inequality.

Proof of Lemma 15. $G(t)=A_{x_0}^{t}$ is differentiable with $G'=f$; MVT on the interval gives θ with $A_{x_0}^{x_0+\mathrm{d}x}=f(\theta)\mathrm{d}x$. □

Proof of Theorem 16. Fix $x_0\in(a,b)$ and $\varepsilon>0$. Let $h\neq 0$ with $x_0+h\in[a,b]$, and let I_h be the closed interval with endpoints x_0 and x_0+h. Since f is continuous at x_0, for $|h|$ sufficiently small there are constants m_ε, M_ε with $m_\varepsilon\leq f(u)\leq M_\varepsilon$ on I_h, $|m_\varepsilon-f(x_0)|<\varepsilon$, $|M_\varepsilon-f(x_0)|<\varepsilon$. By integral monotonicity applied on I_h to the constant functions m_ε and M_ε, for $h>0$ one has $m_\varepsilon h\leq A_{x_0}^{x_0+h}\leq M_\varepsilon h$; for $h<0$ the same inequality holds with both directions reversed (the oriented convention (2) changes the sign of both $A_{x_0}^{x_0+h}$ and h). In either

case, dividing by h preserves the direction of the inequality (both sides change sign together with h), giving

$$m_\varepsilon \;\le\; \frac{A_a^{x_0+h} - A_a^{x_0}}{h} \;\le\; M_\varepsilon,$$

where the numerator uses additivity (5). Since $\varepsilon > 0$ was arbitrary and m_ε, $M_\varepsilon \to f(x_0)$ as $\varepsilon \to 0$, the squeeze yields $\frac{d}{dx} A_a^x|_{x_0} = f(x_0)$; since $x_0 \in (a, b)$ was arbitrary, (29) holds on (a, b), and the one-sided cases at $x = a$, b follow by one-sided limits. □

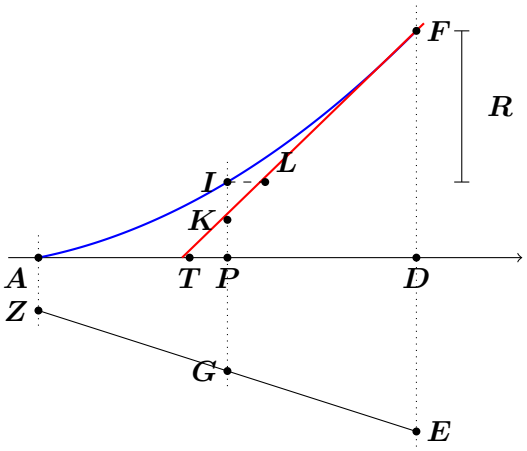


Figure 12. Barrow's Fig. 109.

Proof of Theorem 17. $G(x) = \int_a^x f$ has $G' = f$; so $(F - G)' = 0$ and $F - G$ is constant; $F(b) - F(a) = \int_a^b f$. □

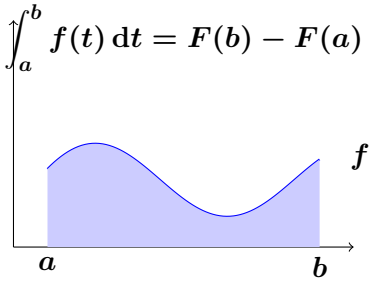


Figure 13. Second FTC.

Proof of Theorem 18. $(fg)' = f'g + fg'$; integrate with Second FTC. □

Proof of Theorem 19. Let $F' = f$. Chain rule: $(F \circ g)' = f(g)g'$; Second FTC on both sides yields (32). □

Proof of Theorem 20. $m \le f \le M$ and $w \ge 0$ give $m \int w \le \int fw \le M \int w$; so $A := \int fw / \int w \in [m, M]$; by IVT $f(c) = A$. □

3.4. The logarithm and the exponential

Proof of Theorem 21. $\ln(ab) - \ln(b) = \int_b^{ab} \mathrm{d}t/t$; the map $(t,\ y) \mapsto (t/b,\ by)$ preserves area and preserves $y = 1/t$, so this equals $\int_1^a \mathrm{d}u/u = \ln a$. □

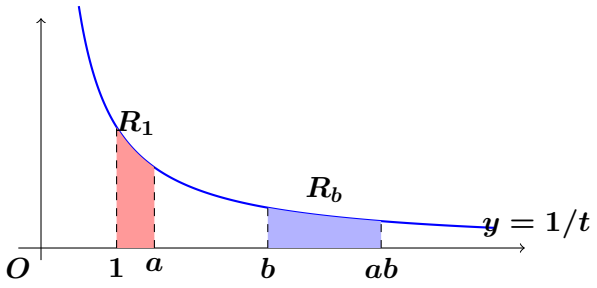


Figure 14. Hyperbolic similarity.

Proof of Theorem 22. $\ln(\exp u \exp v) = \ln(\exp u) + \ln(\exp v) = u + v = \ln(\exp(u+v))$; injectivity of ln gives (35). □

Proof of Theorem 23. Sign-Law gives $\exp(\mathbb{R}_{<0}) =]0,\ 1[, \exp(\{0\}) = \{1\}, \exp(\mathbb{R}_{>0}) =]1,\ \infty[$; Image-Union gives $\exp(\mathbb{R}) = \mathbb{R}_{>0}$. Bijectivity gives $\ln(\mathbb{R}_{>0}) = \mathbb{R}$. □

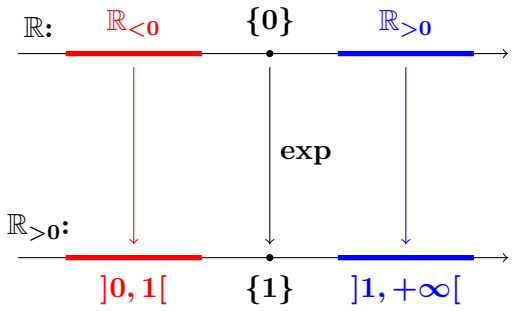


Figure 15. Image of Exponential Function exp

4. Remarks on the geometric development

Barrow's sliding-tangent MVT proof is retained for its historical content; the Barrow-tangency principle (14) is eliminable via Rolle and Monotone + MVT. The FFTC proof uses only area additivity (5) and rectangle comparison (27); no Riemann-sum limit is invoked, as emphasized by Barrow [1] and confirmed in Part II.

Dependency summary

EVT, IVT → Fermat → Rolle → MVT → Const, Monotone, Cauchy, Darboux, Taylor, Convexity, Inverse. Second FTC from FFTC + Const. IBP, Subst, Weighted MVT from Second FTC or IVT/IntMono. Log from IntAdd + Similarity; Exp from Log; Image from LnBij + Sign-Law + ImgUnion. Barrow-area + MVTInt → FFTC → Log.

Part 2. Reverse-Mathematical Analysis in RCA_0

Every Part I entry reappears as a tagged statement with a dependency table and an exact strength in $\{\mathsf{FOL}, \mathsf{RCA}_0, \mathsf{WKL}_0, \mathsf{ACA}_0, \mathsf{ZF}, \mathsf{ZFC}\}$. Geometric proofs are at most WKL_0 as written; §11 gives the matching RCA_0-strength tables via Hardin–Velleman, Yokoyama, or direct argument.

5. Coding conventions

Following Yokoyama [6], we distinguish:

- *Pointwise differentiability:* the limit exists for each $x \in I$; no derivative function needed. Hardin–Velleman's Rolle and MVT in RCA_0 use this.
- *Strong C^k-coding:* $C^k_{\text{strong}}(I)(f) :\iff \exists g_0, \ldots, g_k : \; g_0 = f \wedge \forall j < k, \; g_{j+1} = g_j'$. All C^k hypotheses in Part II mean strong C^k-coding.
- *Inverse function:* Yokoyama [6, Thm. 3.22] in RCA_0 for strong C^1-functions with $f' \neq 0$ on $I, x_0 \in I^\circ$. The pointwise form is not RCA_0-provable.

6. Research context

6.1. The core hierarchy

RCA_0 = recursive comprehension + Σ^0_1-induction; no choice.

Result	Strength	Reference	Choice
MVT	RCA_0	[3]	none
IVT, EVT	WKL_0	[4]	none
sequential IVT/MVT	$\equiv \mathsf{WKL}_0$	[4, 5]	none
ln/exp chain	RCA_0	[6]	none

Key facts:

- MVT $\in \mathsf{RCA}_0$ (Hardin–Velleman [3]); no EVT, no IVT.
- IVT $\equiv \mathsf{WKL}_0$: sole non-RCA_0 ingredient is $\sup\{x : f(x) < 0\}$.
- Inverse function $\in \mathsf{RCA}_0$ (Yokoyama [6, Thm. 3.22]).
- ln/exp $\in \mathsf{RCA}_0$: definitional plus integral postulates.

Remark 1 (Proofs vs. classification).
Part I proofs use sup, inf, *LUB (ax:R-complete). MVT is RCA_0 via Hardin–Velleman. IVT, EVT are WKL_0-equivalent over RCA_0 and remain WKL_0.*

6.2. Illustrative theorems

Theorem	Strength	Reference	Choice
IVT	WKL_0	[4]	none
MVT	RCA_0	[3]	none
Hahn–Banach	$\equiv \mathsf{WKL}_0$	[4]	none
Weierstrass approx.	$\equiv \mathsf{WKL}_0$	[4]	none
Peano existence	$\equiv \mathsf{WKL}_0$	[4]	none
sequential IVT/MVT	$\equiv \mathsf{WKL}_0$	[5]	none
separation	$\equiv \mathsf{ACA}_0$	[15]	none
Banach–Alaoglu	ZFC	[12]	full AC
Tychonoff	ZFC	[13]	full AC
well-ordering	ZFC	[14]	full AC

Strictness: $\mathsf{RCA}_0 \subsetneq \mathsf{WKL}_0 \subsetneq \mathsf{ACA}_0 \subsetneq \mathsf{ZFC}$.

6.3. Dedekind vs. Cauchy completeness

- Dedekind (LUB): every nonempty bounded-above subset has a supremum. A ZF theorem; equivalent over ordered fields to IVT, EVT, MVT, Darboux [7, 8]. Form used here: ax:R-complete.
- Cauchy completeness: every Cauchy sequence converges. ZF-equivalent to $\mathsf{DC}_{\mathbb{R}}$. Not a ZF theorem. Not used here.

Solovay's model [16]: under ZF + DC, all reals are Lebesgue measurable and AC fails; $\mathsf{DC} \Rightarrow \mathsf{DC}_{\mathbb{R}}$, so Cauchy completeness holds there. The choice-free classification is strictly finer than $\mathsf{DC}_{\mathbb{R}}$.

6.4. Constructive RM

In BISH, IVT $\equiv$ LLPO [9]. This document is classical.

6.5. Lean 4 and Mathlib

Lean 4 and Mathlib [10]: about 70.4% of declarations depend on Classical.choice, and 13.1% have choice-free statements with choice-dependent proofs [11]. Our classification is stronger: it certifies choice-free proofs, not just statements.

7. Axioms

The fourteen axioms of $\mathbb{R}$ as a complete ordered field.

Ax 1	ax:add-assoc : $\forall(x,\, y,\, z : \mathbb{R}),\ \ (x + y) + z = x + (y + z)$	Add.Assoc.	ZF
Ax 2	ax:add-comm : $\forall(x,\, y : \mathbb{R}),\ \ x + y = y + x$	Add.Comm.	ZF
Ax 3	ax:add-zero : $\forall(x : \mathbb{R}),\ \ 0 + x = x$	Add.Id.	ZF
Ax 4	ax:add-inv : $\forall(x : \mathbb{R}),\ \ x + (-x) = 0$	Add.Inv.	ZF
Ax 5	ax:mul-assoc : $\forall(x,\, y,\, z : \mathbb{R}),\ \ (x \cdot y) \cdot z = x \cdot (y \cdot z)$	Mul.Assoc.	ZF
Ax 6	ax:mul-comm : $\forall(x,\, y : \mathbb{R}),\ \ x \cdot y = y \cdot x$	Mul.Comm.	ZF

Ax 7	ax:mul-one : $\forall(x:\mathbb{R}),\ 1\cdot x = x$	Mul.Id.	ZF
Ax 8	ax:mul-inv : $\forall(x:\mathbb{R}),$ $(x\neq 0\to x\cdot x^{-1}=1)\wedge(x=0\to x^{-1}=0)$	Mul.Inv.	ZF
Ax 9	ax:distrib : $\forall(x,\,y,\,z:\mathbb{R}),\ x\cdot(y+z)=x\cdot y+x\cdot z$	Distrib.	ZF
Ax 10	ax:nontriv : $0\neq 1$	Nontriv.	ZF
Ax 11	ax:order : $\big(\forall x:\ x\le x\big)\wedge\big(\forall x,$ $y:\ x\le y\wedge y\le x\to x=y\big)\wedge\big(\forall x,\,y,$ $z:\ x\le y\wedge y\le z\to x\le z\big)\wedge\big(\forall x,\,y:\ x\le y\vee y\le x\big)$	LinearOrder	ZF
Ax 12	ax:add-mon : $\forall(x,\,y,\,z:\mathbb{R}),\ x\le y\to x+z\le y+z$	AddMon.	ZF
Ax 13	ax:mul-mon : $\forall(x,\,y:\mathbb{R}),\ 0\le x\wedge 0\le y\to 0\le x\cdot y$	MulMon.	ZF
Ax 14	ax:complete : $\big(\exists x,\,S(x)\big)\wedge\big(\exists u,\,\forall x,$ $S(x)\to x\le u\big)\to\exists s,\,\big(\forall x,\,S(x)\to x\le s\big)\wedge\big(\forall u,\,(\forall x,$ $S(x)\to x\le u)\to s\le u\big)$	Cut	ZF

The Cut axiom enters no Part II proof; it is ZF-realizable via Dedekind cuts without choice.

8. Postulates

Unproved internal statements. Each is RCA_0-provable or first-order logical; none introduces choice.

8.1. Integral postulates

Pos 1	Pos:IntLinear : $\forall(a,b:\mathbb{R}),\ \forall(f,g\in L^1([a,b])),\ \forall(\alpha,\beta:$ $\mathbb{R}),\ \int_a^b(\alpha f(x)+\beta g(x))\,dx=\alpha\int_a^b f(x)\,dx+\beta\int_a^b g(x)\,dx$	IntLin	RCA_0
Pos 2	Pos:IntAdd : $\forall(a,b,c:\mathbb{R}),\ \forall(f\in$ $L^1([\min(a,b,c),\max(a,b,c)])),\ \int_a^b f(x)\,dx=$ $\int_a^c f(x)\,dx+\int_c^b f(x)\,dx$	IntAdd	RCA_0

Note: $f\in L^1([\alpha,\beta])$ and $[\gamma,\delta]\subseteq[\alpha,\beta]$ imply $f\in L^1([\gamma,\delta])$; cited as `restrict` in proof tables.

Pos 3	Pos:IntMono : $\forall(a,b:\mathbb{R}),\ a<b\to\forall(f,g\in$ $L^1([a,b])),\ \big(\forall(x\in[a,b]),\ f(x)\le g(x)\big)\to\int_a^b f(x)\,dx\le$ $\int_a^b g(x)\,dx$	IntMono	RCA_0
Pos 4	Pos:IntSubst : $\forall(a,b:\mathbb{R}),\ \forall(g:\mathbb{R}\to\mathbb{R}),\ \forall(f:\mathbb{R}\to$ $\mathbb{R}),\ g\in C^1([\min(a,b),\max(a,b)])\wedge\big(\forall t\in$ $[\min(a,b),\max(a,b)],\ g'(t)\neq 0\big)\wedge f\in$ $C^0(g([\min(a,b),\max(a,b)]))\to\int_{g(a)}^{g(b)}f(x)\,dx=$ $\int_a^b f(g(t))\,g'(t)\,dt$	IntSubst	RCA_0
Pos 5	Pos:ContInt : $\forall(a,b:\mathbb{R}),\ a<b\to\forall(f:\mathbb{R}\to\mathbb{R}),\ f\in$ $C^0([a,b])\to f\in L^1([a,b])$	ContInt	RCA_0
Pos 6	Pos:Similarity : $\forall(a,b:\mathbb{R}),\ 0<a\wedge 0<b\to\forall(h:\mathbb{R}\to$ $\mathbb{R}),\ \int_b^{a\cdot b}h(t)\,dt=\int_1^a h(bu)\cdot b\,du$	Sim.	RCA_0

8.2. Differential postulates

Pos 7	**Pos:ProdRule** : $\forall(f,g:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(f,x)\wedge\mathsf{Diff}(g,x)\to\mathsf{Diff}(f\cdot g,x)\wedge(f\cdot g)'(x)=f'(x)g(x)+f(x)g'(x)$	**Prod**	$\mathbf{RCA}_0$
Pos 8	**Pos:ChainRule** : $\forall(f,g:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(g,x)\wedge\mathsf{Diff}(f,g(x))\to\mathsf{Diff}(f\circ g,x)\wedge(f\circ g)'(x)=f'(g(x))\cdot g'(x)$	**Chain**	$\mathbf{RCA}_0$
Pos 9	**Pos:InvCont** : $\big(t\mapsto 1/t\big)\in C^0(\mathbb{R}_{>0})$	**InvCont**	$\mathbf{RCA}_0$
Pos 10	**Pos:BarrowTangency** : $\forall(G:\mathbb{R}\to\mathbb{R}),\ \forall(x_0,m:\mathbb{R}),\ \mathsf{Diff}(G,x_0)\wedge\Big(\big(\forall(u\neq x_0)\text{ near }x_0,\ G(x_0)+m(u-x_0)\le G(u)\big)\vee\big(\forall(u\neq x_0)\text{ near }x_0,\ G(x_0)+m(u-x_0)\ge G(u)\big)\Big)\to m=G'(x_0)$	**Barrow**	$\mathbf{RCA}_0$

8.3. Logarithmic postulates

Pos 11	**Pos:LnBij** : $\mathsf{Bijective}(\ln:\mathbb{R}_{>0}\to\mathbb{R})$	**LnBij**	$\mathbf{RCA}_0$
Pos 12	**Pos:LnSign** : $\forall(a:\mathbb{R}_{>0}),\ \big(\ln(a)<0\leftrightarrow a<1\big)\wedge\big(\ln(a)=0\leftrightarrow a=1\big)\wedge\big(\ln(a)>0\leftrightarrow a>1\big)$	**LnSign**	$\mathbf{RCA}_0$

8.4. Logical postulate

Pos 13	**Pos:ImgUnion** : $\forall(X,Y:\mathsf{Type}),\ \forall(f:X\to Y),\ \forall(A,B:X\to\mathsf{Prop}),\ f(A\cup B)=f(A)\cup f(B)$	**ImgUnion**	**FOL**

8.5. External results

Pos 14	**Pos:ContImgBdd** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\big(\exists L,U:\mathbb{R},\ \forall x:[a,b],\ L\le f(x)\le U\big)$	**BddImg**	$\mathbf{RCA}_0$
Pos 15	**Pos:ContImgAttain** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\big(\exists x_{\max}:[a,b],\forall x:[a,b],f(x)\le f(x_{\max})\big)\wedge\big(\exists x_{\min}:[a,b],\forall x:[a,b],f(x_{\min})\le f(x)\big)$	**Attain**	$\mathbf{WKL}_0$

Note on Pos:ContImgBdd and the coding of $C^0([a,b])$. Under the coding of §5 (following Simpson [4]), a continuous function $f:[a,b]\to\mathbb{R}$ carries a coded modulus of continuity, from which boundedness of the image $f([a,b])$ follows in $\mathbf{RCA}_0$. The **Pos:ContImgBdd** tag $\mathbf{RCA}_0$ reflects this standard coding convention. Under an alternative coding — pointwise continuity without a coded modulus — boundedness of continuous functions on $[a,b]$ rises to $\mathbf{WKL}_0$; the choice between codings is a foundation choice, not a mathematical error, and §5 fixes the Simpson convention for this document.

Pos 16	**Pos:Π^0_1Sup** : $\forall(\varphi:\mathbb{N}\times\mathbb{R}\to\{0,1\}),\ \forall(a,b:\mathbb{R}),\ a\le b\to\big(\exists x\in[a,b],\forall n\,\varphi(n,x)=0\big)\to\exists s\in[a,b],\ \big(\forall x\in[a,b],(\forall n\,\varphi(n,x)=0)\to x\le s\big)\wedge\big(\forall u\in[a,b],(\forall x\in[a,b],(\forall n\,\varphi(n,x)=0)\to x\le u)\to s\le u\big)$	**Sup**	$\mathbf{WKL}_0$
Pos 17	**Pos:Σ^0_1Choice** : $\big(\forall n,\exists m,R(n,m)\big)\to\exists g,\forall n,R(n,g(n))$	Σ^0_1	$\mathbf{RCA}_0$

Pos 18	**Pos:SeqCont** : $\forall(f:[a,b]\to\mathbb{R}), f\in C^0([a,b])\to\forall x:[a,b],\ \forall(x_n:\mathbb{N}\to[a,b]),\ (x_n\to x)\to\big(f(x_n)\to f(x)\big)$	**SeqCont**	$\mathbf{RCA_0}$
Pos 19	**Pos:HVRolle** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge\big(\forall x:(a,b),\ \mathsf{Diff}(f,x)\big)\wedge f(a)=f(b)\to\exists c:(a,b),\ f'(c)=0$	**HVRolle**	$\mathbf{RCA_0}$
Pos 20	**Pos:YokoyamaInv** : $\forall(f:I\to\mathbb{R}),\ \forall(x_0:I^\circ),\ C^1_{\mathbf{strong}}(I)(f)\wedge(\forall x:I, f'(x)\neq 0)\wedge\mathsf{StrictMono}(f)\to\mathsf{Diff}(f^{-1},f(x_0))\wedge(f^{-1})'(f(x_0))=1/f'(x_0)$	**YokInv**	$\mathbf{RCA_0}$

9. Definitions

Definitional abbreviations available in $\mathbf{RCA_0}$: set notation by $\mathbf{\Delta^0_1}$-comprehension; L^1, C^k, $C^k_{\mathbf{strong}}$ by $\mathbf{\Sigma^0_1}$ predicates; A^x_a, ln, exp by unique existence + graphs; Diff, ChordSlope, TangentSlope, LocalExt, Increasing, StrictInc, StrictDec, StrictMono by definitional extension.

Def 1	**Def:R$_{>0}$** : $\mathbb{R}_{>0}:=\{x:\mathbb{R}\mid x>0\},\ \mathbb{R}_{<0}:=\{x:\mathbb{R}\mid x<0\},\ \mathbb{R}_{\geq 0}:=\{x:\mathbb{R}\mid x\geq 0\}$	**R$_{>0}$**	$\mathbf{RCA_0}$
Def 2	**Def:L^1** : $L^1([a,b]):=\{f:\mathbb{R}\to\mathbb{R}\mid\int_a^b|f(x)|\,dx<\infty\}$	**L^1**	$\mathbf{RCA_0}$

Def 3	**Def:C^k** : $\forall(k:\mathbb{N}),\ \forall(U:\mathbb{R}\to\mathsf{Prop}),\ C^k(U):=\{f:\mathbb{R}\to\mathbb{R}\mid f\ \mathsf{k-timescont.\ diff.\ on}\ U\}$	**C^k**	$\mathbf{RCA_0}$
Def 4	**Def:C$^k_{\mathbf{strong}}$** : $\forall(k:\mathbb{N}),\ \forall(I:\mathbb{R}\to\mathsf{Prop}),\ C^k_{\mathbf{strong}}(I)(f):\iff\exists(g_0,\dots,g_k:\mathbb{R}\to\mathbb{R}),\ g_0=f\wedge\big(\forall j\leq k,\ g_j\in C^0(I)\big)\wedge\big(\forall j<k,\ g_{j+1}=(g_j)'\big)$	**Strong C^k**	$\mathbf{RCA_0}$
Def 5	**Def:Diff** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(f,x):\iff\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ exists; $f'(x):=$ that limit	**Diff**	$\mathbf{RCA_0}$
Def 6	**Def:Area** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a,x:\mathbb{R}),\ A^x_a:=\int_a^x f(t)\,dt$ with $A^x_x=0,\ A^{x+\mathrm{d}x}_x=-A^x_{x+\mathrm{d}x}\ (\mathrm{d}x<0)$	**Area**	$\mathbf{RCA_0}$
Def 7	**Def:chord** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a,b:\mathbb{R}),\ a\neq b\to\mathsf{ChordSlope}(f,a,b):=\frac{f(b)-f(a)}{b-a}$	**Chord**	$\mathbf{RCA_0}$
Def 8	**Def:tangent** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(c:\mathbb{R}),\ \mathsf{Diff}(f,c)\to\mathsf{TangentSlope}(f,c):=f'(c)$	**Tangent**	$\mathbf{RCA_0}$
Def 9	**Def:LocExt** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(c:\mathbb{R}),\ \mathsf{LocalExt}(f,c):\iff\exists(\varepsilon>0),\ \big(\forall(x\in(c-\varepsilon,c+\varepsilon)),\ f(x)\leq f(c)\big)\vee\big(\forall(x\in(c-\varepsilon,c+\varepsilon)),\ f(x)\geq f(c)\big)$	**LocExt**	$\mathbf{RCA_0}$
Def 10	**Def:Increasing** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{Increasing}(f):\iff\forall(x,y:\mathbb{R}),\ x<y\to f(x)\leq f(y)$	**Incr**	$\mathbf{RCA_0}$
Def 11	**Def:StrictInc** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictInc}(f):\iff\forall(x,y:\mathbb{R}),\ x<y\to f(x)<f(y)$	**SIncr**	$\mathbf{RCA_0}$
Def 12	**Def:StrictDec** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictDec}(f):\iff\forall(x,y:\mathbb{R}),\ x<y\to f(x)>f(y)$	**SDecr**	$\mathbf{RCA_0}$
Def 13	**Def:StrictMono** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictMono}(f):\iff\mathsf{StrictInc}(f)\vee\mathsf{StrictDec}(f)$	**SMono**	$\mathbf{RCA_0}$
Def 14	**Def:ln** : $\forall(x:\mathbb{R}_{>0}),\ \ln(x):=\int_1^x\frac{1}{t}\,dt$	**ln**	$\mathbf{RCA_0}$
Def 15	**Def:exp** : $\exp:=\ln^{-1}:\mathbb{R}\to\mathbb{R}_{>0}$	**exp**	$\mathbf{RCA_0}$

10. Statements

Statements and their strengths. Proofs in §11. All C^k hypotheses are strong C^k-coding (§5); pointwise differentiability appears only in Hardin–Velleman.

10.1. Topological theorems

Thm 1	**EVT** : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \to \exists x_{\max} \in [a,b],\ \exists x_{\min} \in [a,b],\ \forall (x \in [a,b]),\ f(x_{\min}) \le f(x) \le f(x_{\max})$	EVT	$\mathbf{WKL_0}$
Thm 2	**IVT** : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f(a) \cdot f(b) < 0 \to \exists (c \in (a,b)),\ f(c) = 0$	IVT	$\mathbf{WKL_0}$

10.2. Differentiation theorems

Thm 3	**Fermat** : $\forall (f : \mathbb{R} \to \mathbb{R}),\ \forall (a : \mathbb{R}),\ \forall (b : \mathbb{R}),\ \forall (c : \mathbb{R}),\ a < c \wedge c < b \wedge \mathsf{LocalExt}(f,c) \wedge \mathsf{Diff}(f,c) \to f'(c) = 0$	Fermat	$\mathbf{RCA_0}$
Thm 4	**Rolle** : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f \in C^1((a,b)) \wedge f(a) = f(b) \to \exists (c \in (a,b)),\ f'(c) = 0$	Rolle	$\mathbf{RCA_0}$
Thm 5	**MVT** : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f \in C^1((a,b)) \to \exists (c \in (a,b)),\ f'(c) = \mathsf{ChordSlope}(f,a,b)$	MVT	$\mathbf{RCA_0}$
Thm 6	**Const** : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f \in C^1((a,b)) \wedge (\forall x \in (a,b),\ f'(x) = 0) \to (\forall x \in [a,b],\ f(x) = f(a))$	Const	$\mathbf{RCA_0}$
Thm 7	**Monotone** : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f \in C^1((a,b)) \to \big((\forall x \in (a,b),\ f'(x) \ge 0) \to \mathsf{Increasing}(f)\big) \wedge \big((\forall x \in (a,b),\ f'(x) > 0) \to \mathsf{StrictInc}(f)\big)$	Mono	$\mathbf{RCA_0}$
Thm 8	**CauchyMVT** : $\forall (f,g : [a,b] \to \mathbb{R}),\ f,g \in C^0([a,b]) \wedge f,g \in C^1((a,b)) \wedge (\forall x \in (a,b),\ g'(x) \neq 0) \to \exists (c \in (a,b)),\ \frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$	Cauchy	$\mathbf{RCA_0}$
Thm 9	**Darboux** : $\forall (f : [a,b] \to \mathbb{R}),\ \forall (\lambda : \mathbb{R}),\ \mathsf{Diff}(f,a) \wedge \mathsf{Diff}(f,b) \wedge \big(\forall x \in (a,b),\ \mathsf{Diff}(f,x)\big) \wedge f'(a) < \lambda < f'(b) \to \exists (c \in (a,b)),\ f'(c) = \lambda$	Darboux	$\mathbf{WKL_0}$
Thm 10	**LHopital** : $\forall (f,g : \mathbb{R} \to \mathbb{R}),\ \forall (a : \mathbb{R}),\ \forall (L : \mathbb{R}),\ f(a) = g(a) = 0 \wedge \big(\forall x \text{ near } a,\ \mathsf{Diff}(f,x) \wedge \mathsf{Diff}(g,x) \wedge g'(x) \neq 0\big) \wedge \lim_{x \to a} \frac{f'(x)}{g'(x)} = L \to \lim_{x \to a} \frac{f(x)}{g(x)} = L$	L'Hôpital	$\mathbf{RCA_0}$
Thm 11	**Taylor** : $\forall (f : \mathbb{R} \to \mathbb{R}),\ \forall (a,x : \mathbb{R}),\ \forall (n : \mathbb{N}),\ a \neq x \wedge C^{n+1}_{\mathrm{strong}}([\min(a,x), \max(a,x)])(f) \to \exists (c \in (\min(a,x), \max(a,x))),\ f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$	Taylor	$\mathbf{RCA_0}$
Thm 12	**Convexity** : $\forall (f : [a,b] \to \mathbb{R}),\ C^2_{\mathrm{strong}}([a,b])(f) \wedge (\forall x \in (a,b),\ f''(x) \ge 0) \to \forall (x, x_0 \in [a,b]),\ f(x) \ge f(x_0) + f'(x_0)(x - x_0)$	Convex	$\mathbf{RCA_0}$
Thm 13	**Inverse** : $\forall (f : I \to \mathbb{R}),\ \forall (x_0 \in I^\circ),\ C^1_{\mathrm{strong}}(I)(f) \wedge (\forall x \in I,\ f'(x) \neq 0) \wedge \mathsf{StrictMono}(f) \to \mathsf{Diff}(f^{-1}, f(x_0)) \wedge (f^{-1})'(f(x_0)) = 1/f'(x_0)$	Inv	$\mathbf{RCA_0}$

10.3. Integration theorems

Lem 1	**BarrowArea** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge\mathsf{Increasing}(f)\to\forall(x_0\in[a,b]),\ \forall(\mathrm{d}x\neq 0,\ x_0+\mathrm{d}x\in[a,b]),\ f(x_0)\,\mathrm{d}x\le A_{x_0}^{x_0+\mathrm{d}x}\le f(x_0+\mathrm{d}x)\,\mathrm{d}x$	BarrowArea	RCA_0
Lem 2	**MVTInt** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\forall(x_0,\mathrm{d}x\neq 0),\ x_0+\mathrm{d}x\in[a,b]\to\exists(\theta\text{ between }x_0,x_0+\mathrm{d}x),\ A_{x_0}^{x_0+\mathrm{d}x}=f(\theta)\,\mathrm{d}x$	MVTInt	RCA_0
Thm 14	**FFTC** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\forall(x\in(a,b)),\ \frac{d}{dx}A_a^x=f(x)$	FFTC	RCA_0
Thm 15	**SecondFTC** : $\forall(F:[a,b]\to\mathbb{R}),\ F\in C^1([a,b])\to\int_a^b F'(t)\,dt=F(b)-F(a)$	SecondFTC	RCA_0
Thm 16	**IBP** : $\forall(f,g\in C^1([a,b])),\ \int_a^b f\,g'=[fg]_a^b-\int_a^b f'\,g$	IBP	RCA_0
Thm 17	**Subst** : $\forall(g\in C^1([a,b])),\ \forall(f\in C^0(g([a,b]))),\ \int_{g(a)}^{g(b)}f(u)\,du=\int_a^b f(g(t))\,g'(t)\,dt$	Subst	RCA_0
Thm 18	**WeightedMVT** : $\forall(f\in C^0([a,b])),\ \forall(w\ge 0\text{ Riemann integrable}),\ \int_a^b w>0\to\exists(c\in[a,b]),\ \int_a^b f\,w=f(c)\int_a^b w$	WMVT	WKL_0

10.4. The logarithm and the exponential

Lem 3	**PosClosedMul** : $\forall(a,b:\mathbb{R}),\ 0<a\wedge 0<b\to 0<a\cdot b$	Pos Closed	RCA_0
Lem 4	**LogProperty** : $\forall(a,b:\mathbb{R}_{>0}),\ \ln(a)+\ln(b)=\ln(a\cdot b)$	LogAdd	RCA_0
Lem 5	**ExpProperty** : $\forall(u,v:\mathbb{R}),\ \exp(u)\cdot\exp(v)=\exp(u+v)$	ExpAdd	RCA_0
Thm 19	**ImageExpLn** : $\exp(\mathbb{R})=\mathbb{R}_{>0}\wedge\ln(\mathbb{R}_{>0})=\mathbb{R}$	Image	RCA_0

10.5. Illustrative theorems

Theorem	Strength	Choice
IVT	WKL_0	none
MVT	RCA_0	none
Hahn–Banach	WKL_0	none
Weierstrass	WKL_0	none
Peano	WKL_0	none
separation	ACA_0	none
Banach–Alaoglu	ZFC	full AC
Tychonoff	ZFC	full AC

Remark 2 (EVT tag).

EVT $\equiv$ ***WKL***$_0$ *over* ***RCA***$_0$ [4]. *Entries whose printed proofs invoke EVT transitively: Rolle, MVT, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity, Darboux, WeightedMVT; Part I proofs are at most* ***WKL***$_0$ *as written. Rolle and MVT are* ***RCA***$_0$ *via Hardin–Velleman* [3]*; the descendants inherit* ***RCA***$_0$ *under strong-*C^k*-coding. Darboux and WeightedMVT are* ***WKL***$_0$ *as stated. Fermat,*

FFTC, SecondFTC, IBP, Subst, BarrowArea, MVTInt, Inverse, and the ln/exp *chain are* RCA_0 *(direct;* [3, 6] *for Inverse and MVTInt). IVT is* WKL_0*-equivalent for the* sup *form; the neighbourhood argument below uses only continuity.*

11. Proofs

Starred subsections, one per proof, in statement order. Each proof is a four-column dependency table: first row Hyp, last row $\to$ I$(1, \dots, N-1)$. Justifications use only Hyp, def, intro, Ax:, Pos:, Def:, Lem:, Thm:, Eq$(\dots)$, or a logical rule ($\to$ I, $\wedge$E, $\vee$E, $\forall$I, $\forall$E, $\exists$I, $\exists$E, $\neg$I, $\neg$E, $\bot$I, $\bot$E, RAA, DNE, LEM). No external citation appears inside a proof table.

Proof of Theorem 1 (EVT)

1	$f \in C^0([a,\,b])$	hypothesis	Hyp
2	$\exists L,\,U,\ \forall x,\ L \le f(x) \le U$	bdd	Pos:14(1)
3	$\exists x_{\max},\ \forall x,\ f(x) \le f(x_{\max})$	max	Pos:15(1)
4	$\exists x_{\min},\ \forall x,\ f(x_{\min}) \le f(x)$	min	Pos:15(1)
5	$\forall x,\ f(x_{\min}) \le f(x) \le f(x_{\max})$	combine	Eq(3, 4)
6	$\top$	close	$\to$ I$(1, \dots, 5)$

Proof of Theorem 2 (IVT)

1	$f \in C^0([a,\,b]) \wedge f(a)f(b) < 0$	hypothesis	Hyp
2	$f(a) < 0 < f(b)$	sign	Eq(1)
3	$S := \{x \in [a,b] : f(x) < 0\}$ is Σ^0_1	def	Eq(2)
4	$P := \{x \in [a,b] : -f(x) \le 0\}$ is Π^0_1	def	Eq(2)
5	$c := \sup S = -\inf P$	sup	Pos:16(4)
6	$\forall n\, \exists x \in S,\ c - \frac{1}{n+1} < x \le c$	sup	Eq(5)
7	$\exists (x_n) \subset S,\ x_n \to c$ (codes of reals)	Σ^0_1-choice	Pos:17(6)
8	$f(x_n) \to f(c)$	seq	Pos:18(1, 7)
9	$f(c) \le 0$	lim	Eq(7, 8)
10	$f(c) < 0 \to \exists \delta,\ \forall u \in (c-\delta,\, c+\delta),\ f(u) < 0$	cont	Def:3(1)
11	$f(c) < 0 \to (c + \delta/2 \in S \wedge c < c + \delta/2)$	shift	Eq(3, 10)
12	$f(c) < 0 \to \bot$	contra	Eq(5, 11)
13	$f(c) \ge 0$	RAA	Eq(12)
14	$f(c) = 0$	antisym	Eq(9, 13)
15	$\top$	close	$\to$ I$(1, \dots, 14)$

Coding note (IVT). In row 3, $S = \{x \in [a,b] : f(x) < 0\}$ is read as a Σ^0_1 set of *codes* for reals, so Pos:Σ^0_1Choice (row 7) applies at the level of codes. In row 4, $P = \{x \in [a,b] : -f(x) \le 0\}$ is the Π^0_1 complement of S in $[a,b]$; Pos:Π^0_1Sup applied to P yields $\inf P$, and $\sup S = -\inf P$ (cf. Simpson [4, §IV.2]).

Proof of Theorem 3 (Fermat)

1	local max, $\mathsf{Diff}(f, c)$	hypothesis	Hyp
2	$\forall h > 0$ small, $(f(c+h) - f(c))/h \leq 0$	locmax	Def:9(1)
3	$f'(c) \leq 0$	osl	Def:5(2)
4	$\forall h < 0$ small, $(f(c+h) - f(c))/h \geq 0$	locmax	Def:9(1)
5	$f'(c) \geq 0$	osl	Def:5(4)
6	$f'(c) = 0$	antisym	Ax:11(3, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 4 (Rolle)

1	$f \in C^0 \wedge f \in C^1 \wedge f(a) = f(b)$	hypothesis	Hyp
2	$\exists c,\ f'(c) = 0$	HVRolle	Pos:19(1)
3	$\top$	close	$\to$ I(1, 2)

Proof of Theorem 5 (MVT)

1	$f \in C^0 \wedge f \in C^1$	hypothesis	Hyp
2	$L :=$ chord, $g := f - L$	def	Def:7(1)
3	$g \in C^0 \wedge g \in C^1 \wedge g(a) = g(b) = 0$	construct	Eq(2)
4	$\exists c,\ g'(c) = 0$	HVRolle	Pos:19(3)
5	$g'(c) = f'(c) -$ ChordSlope	diff	Eq(2)
6	$f'(c) =$ ChordSlope	combine	Eq(4, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 6 (Const)

1	$f \in C^0 \wedge f \in C^1 \wedge f' \equiv 0$	hypothesis	Hyp
2	$\forall x,\ \exists c,\ f(x) - f(a) = f'(c)(x - a)$	MVT	Thm:5(1)
3	$f'(c)(x - a) = 0$	combine	Eq(1, 2)
4	$f(x) = f(a)$	combine	Eq(2, 3)
5	$\top$	close	$\to$ I(1, ..., 4)

Proof of Theorem 7 (Monotone)

1	$f \in C^0 \wedge f \in C^1$	hypothesis	Hyp
2	$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$	MVT	Thm:5(1)
3	$f' \geq 0 \to f(x_2) \geq f(x_1)$	sign	Eq(1, 2)
4	$f' > 0 \to f(x_2) > f(x_1)$	sign	Eq(1, 2)
5	$f' \geq 0 \to$ Increasing	combine	Def:10(3)
6	$f' > 0 \to$ StrictInc	combine	Def:11(4)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 8 (CauchyMVT)

1	$f, g \in C^0 \wedge f, g \in C^1 \wedge g' \neq 0$	hypothesis	Hyp
2	$h := (f(b) - f(a))g - (g(b) - g(a))f$	def	def
3	$h \in C^0 \wedge h \in C^1 \wedge h(a) = h(b)$	construct	Eq(2)
4	$\exists c,\ h'(c) = 0$	HVRolle	Pos:19(3)
5	$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$	diff	Eq(2, 4)
6	$g'(c) \neq 0$	hyp	Eq(1, 4)
7	$g(b) \neq g(a)$	Rolle(g)	Thm:4(1)
8	$f'(c)/g'(c) = (f(b) - f(a))/(g(b) - g(a))$	divide	Eq(5, 6, 7)
9	$\top$	close	$\to$ I(1, ..., 8)

Proof of Theorem 9 (Darboux)

1	f' exists, $f'(a) < \lambda < f'(b)$	hypothesis	Hyp
2	$g := f - \lambda x,\ g'(a) < 0 < g'(b)$	def	Eq(1)
3	$\exists h > 0,\ g(a + h) < g(a)$	$g'(a) < 0$	Def:5(2)
4	$\exists h > 0,\ g(b - h) < g(b)$	$g'(b) > 0$	Def:5(2)
5	$\exists c \in [a, b],\ \forall x,\ g(c) \leq g(x)$	EVT	Thm:1(1)
6	$c \neq a \wedge c \neq b$	combine	Eq(3, 4, 5)
7	$c \in (a, b)$	combine	Eq(5, 6)
8	$g'(c) = 0$	Fermat	Thm:3(7)
9	$f'(c) = \lambda$	def	Eq(2, 8)
10	$\top$	close	$\to$ I(1, ..., 9)

Proof of Theorem 10 (LHopital)

1	$f(a) = g(a) = 0 \wedge (\forall x \text{ near } a,\ g'(x) \neq 0) \wedge \lim_{x \to a} f'/g' = L$	hypothesis	Hyp
2	$g \neq 0$ near a	Rolle	Thm:4(1)
3	$f(x)/g(x) = f'(c_x)/g'(c_x)$	Cauchy	Thm:8(1, 2)
4	$x \to a^+ \Rightarrow c_x \to a^+$	combine	Eq(3)
5	$f'/g'(c_x) \to L$	combine	Eq(1, 4)
6	$f/g(x) \to L\ (a^+)$	combine	Eq(3, 5)
7	$f/g(x) \to L\ (a^-)$ sym.	Cauchy	Thm:8(1, 2)
8	$\lim_{x \to a} f/g = L$	combine	Eq(6, 7)
9	$\top$	close	$\to$ I(1, ..., 8)

Proof of Theorem 11 (Taylor)

1	$f \in C^{n+1}$	hypothesis	Hyp
2	$P :=$ Taylor poly	def	def

3	$M := (f(x) - P(x))/(x-a)^{n+1}$	**def**	**Eq(1, 2)**
4	$h := f - P - M(t-a)^{n+1}$	**def**	**Eq(2, 3)**
5	$h^{(j)}(a) = 0 \ (j \le n)$	**def**	**Eq(2, 4)**
6	$h(x) = 0$	**def**	**Eq(3, 4)**
7	$h \in C^n \wedge h \in C^{n+1}$	**smooth**	**Eq(1, 4)**
8	$\exists c_1,\ h'(c_1) = 0$	**HVRolle**	**Pos:19(7)**
9	$h'(a) = h'(c_1) = 0$	**combine**	**Eq(5, 8)**
10	$\exists c_2,\ h''(c_2) = 0$	**HVRolle**	**Pos:19(7)**
11	(iterate $n+1$ times)	**HVRolle**	**Pos:19(7)**
12	$\exists c,\ h^{(n+1)}(c) = 0$	**combine**	**Eq(8, ..., 11)**
13	$h^{(n+1)} = f^{(n+1)} - M(n+1)!$	**def**	**Eq(2, 4)**
14	$M = f^{(n+1)}(c)/(n+1)!$	**combine**	**Eq(12, 13)**
15	$f(x) = \text{Taylor sum} + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$	**combine**	**Eq(2, 3, 14)**
16	$\top$	**close**	**$\to$ I(1, ..., 15)**

Proof of Theorem 12 (Convexity)

1	$f \in C^2 \wedge f'' \ge 0$	**hypothesis**	**Hyp**
2	$g := f - f(x_0) - f'(x_0)(x - x_0)$	**def**	**Def:8(1)**
3	$g'' = f'' \ge 0$	**diff**	**Eq(1, 2)**
4	g' nondecreasing, $g'(x_0) = 0$	**Monotone**	**Thm:7(3)**
5	$g' \le 0 \ (x < x_0),\ g' \ge 0 \ (x > x_0)$	**split**	**Eq(4)**
6	g nonincr./nondecr.	**Monotone**	**Thm:7(5)**
7	$g(x_0) = 0$	**def**	**Eq(2)**
8	$g \ge 0$	**combine**	**Eq(6, 7)**
9	$f(x) \ge f(x_0) + f'(x_0)(x - x_0)$	**combine**	**Eq(2, 8)**
10	$\top$	**close**	**$\to$ I(1, ..., 9)**

Proof of Theorem 13 (Inverse)

1	$x_0 \in I^\circ \wedge C^1_{\text{strong}}(I)(f) \wedge (\forall x : I,\ f'(x) \neq 0) \wedge \mathsf{StrictMono}(f)$	**hypothesis**	**Hyp**
2	$\mathsf{Diff}(f^{-1}, f(x_0)) \wedge (f^{-1})'(f(x_0)) = 1/f'(x_0)$	**Yokoyama**	**Pos:20(1)**
3	$\top$	**close**	**$\to$ I(1, 2)**

Proof of Lemma 1 (BarrowArea)

1	$f \in C^0([a,b]),\ \mathsf{Incr}(f),\ x_0 < x_0 + \mathrm{d}x$	**hypothesis**	**Hyp**
2	$f \in L^1([a,b])$	**cont**	**Pos:5(1)**
3	$f(x_0) \le f \le f(x_0 + \mathrm{d}x)$	**incr**	**Def:10(1)**
4	inscribed $\le A \le$ circumscribed	**mono**	**Pos:3(2,3)**

5	$f(x_0)\mathrm{d}x \le A \le f(x_0+\mathrm{d}x)\mathrm{d}x$	rect	Eq(4)
6	$\mathrm{d}x < 0:\ A = -A'$	orient	Def:6
7	same inequality	negate	Eq(6)
8	$\top$	close	$\to$ I(1, ..., 7)

Proof of Lemma 2 (MVTInt)

1	$f \in C^0,\ x_0, x_0+\mathrm{d}x \in [a,b]$	hypothesis	Hyp
2	$G(t) := A_{x_0}^t$	area	Def:6(1)
3	$G \in C^0 \wedge G' = f$	FFTC	Thm:14(1)
4	$\exists\theta,\ G'(\theta) = (G(x_0+\mathrm{d}x) - G(x_0))/\mathrm{d}x$	MVT	Thm:5(1,3)
5	$G(x_0) = 0 \wedge G(x_0+\mathrm{d}x) = A$	def	Def:6(2)
6	$A = f(\theta)\mathrm{d}x$	combine	Eq(3, 4, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 14 (FFTC)

1	$f \in C^0,\ x_0 \in (a,b),\ \varepsilon > 0$	hypothesis	Hyp
2	$f \in L^1([a,b])$	cont	Pos:5(1)
3	$f \in L^1([\min(a,x_0,x_0+h), \max(a,x_0,x_0+h)])$	restrict	Eq(2)
4	$A_a^{x_0+h} = A_a^{x_0} + A_{x_0}^{x_0+h}$	additivity	Pos:2(3)
5	$\exists m_\varepsilon, M_\varepsilon,\ \forall u \in I_h,\ m_\varepsilon \le f(u) \le M_\varepsilon,\ \lvert m_\varepsilon - f(x_0)\rvert,$ $\lvert M_\varepsilon - f(x_0)\rvert < \varepsilon$	cont	Def:3(1)
6	$h > 0:\ m_\varepsilon h \le A \le M_\varepsilon h$	mono	Pos:3(3,5)
7	$h < 0:\ M_\varepsilon h \le A \le m_\varepsilon h$	orient	Def:6
8	$m_\varepsilon \le (A_a^{x_0+h} - A_a^{x_0})/h \le M_\varepsilon$	divide	Eq(4, 6, 7)
9	$\mathrm{d}A_a^x/\mathrm{d}x\rvert_{x_0} = f(x_0)$	squeeze	Eq(5, 8)
10	x_0 arbitrary	gen	Eq(9)
11	$\top$	close	$\to$ I(1, ..., 10)

Proof of Theorem 15 (SecondFTC)

1	$F \in C^1,\ f := F' \in C^0$	hypothesis	Hyp
2	$G(x) := \int_a^x f,\ G' = f$	FFTC	Thm:14(1)
3	$(F-G)' = 0$	diff	Eq(1, 2)
4	$F - G$ constant	Const	Thm:6(3)
5	$F(b) - F(a) = \int_a^b f$	combine	Eq(4)
6	$\top$	close	$\to$ I(1, ..., 5)

Proof of Theorem 16 (IBP)

1	$f,\, g \in C^1$	hypothesis	Hyp
2	$f'g,\, fg' \in C^0([a,\, b]) \subseteq L^1([a,\, b])$	cont	Pos:5(1)
3	$(fg)' = f'g + fg'$	prod	Pos:7(1)
4	$\int_a^b (fg)' = [fg]_a^b$	SecondFTC	Thm:15(2)
5	$\int (f'g + fg') = \int f'g + \int fg'$	linear	Pos:1(2, 4)
6	$[fg]_a^b = \int f'g + \int fg'$	combine	Eq(4, 5)
7	$\int fg' = [fg]_a^b - \int f'g$	rearrange	Eq(6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Theorem 17 (Subst)

1	$g \in C^1,\ \ f \in C^0(g([a,\, b]))$	hypothesis	Hyp
2	$f \in L^1(g([a,\, b])) \wedge f(g)g' \in L^1([a,\, b])$	cont	Pos:5(1)
3	$F := \int_{g(a)}^{\cdot} f,\ \ F' = f$	FFTC	Thm:14(1)
4	$(F \circ g)' = f(g)g'$	chain	Pos:8(1, 3)
5	$\int f(g)g' = F(g(b)) - F(g(a))$	SecondFTC	Thm:15(4)
6	$\int_{g(a)}^{g(b)} f = F(g(b)) - F(g(a))$	SecondFTC	Thm:15(3)
7	both $= \ F(g(b)) - F(g(a))$	combine	Eq(5, 6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Theorem 18 (WeightedMVT)

1	$f \in C^0,\ \ w \geq 0$ int., $\int w > 0$	hypothesis	Hyp
2	$f,\, fw \in L^1([a,\, b])$	int	Pos:5(1)
3	$m := \min f,\ \ M := \max f$	EVT	Thm:1(1)
4	$m \int w \leq \int fw \leq M \int w$	mono	Pos:3(2, 3)
5	$A := \int fw / \int w \in [m,\, M]$	divide	Eq(4)
6	$\exists c,\ \ f(c) = A$	IVT	Thm:2(1, 5)
7	$\int fw = f(c) \int w$	combine	Eq(5, 6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Lemma 3 (PosClosedMul)

1	$0 < a,\ \ 0 < b$	hypothesis	Hyp
2	$0 \leq a,\ \ 0 \leq b$	weaken	Ax:11(1)
3	$0 \leq ab$	mulMon	Ax:13(2)
4	$a \neq 0,\ \ b \neq 0$	combine	Ax:11(2)
5	$ab \neq 0$	combine	Ax:8(4)
6	$0 < ab$	combine	Eq(3, 5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Lemma 4 (LogProperty)

1	$a,\, b > 0$	hypothesis	Hyp
2	$ab > 0$	closed	Lem:3(1)
3	$(t \mapsto 1/t) \in C^0(\mathbb{R}_{>0})$	InvCont	Pos:9
4	$(t \mapsto 1/t) \in L^1([\min(1,b,ab), \max(1,b,ab)])$	ContInt	Pos:5(3)
5	$\int_1^{ab} \frac{\mathrm{d}t}{t} = \int_1^{b} \frac{\mathrm{d}t}{t} + \int_b^{ab} \frac{\mathrm{d}t}{t}$	additivity	Pos:2(4)
6	$\ln(ab) = \int_1^{ab} \frac{\mathrm{d}t}{t},\ \ \ln(b) = \int_1^{b} \frac{\mathrm{d}t}{t}$	def	Def:14(1,2)
7	$\ln(ab) - \ln(b) = \int_b^{ab} \frac{\mathrm{d}t}{t}$	combine	Eq(5, 6)
8	$g := (u \mapsto bu) \in C^1([\min(1,a), \max(1,a)]),\ \ g' = b \neq 0$	lin	Ax:6(1,2)
9	$\int_b^{ab} \frac{\mathrm{d}t}{t} = \int_1^{a} \frac{1}{bu}\, b\, \mathrm{d}u$	subst	Pos:4(3,8)
10	$\frac{1}{bu} \cdot b = \frac{1}{u}$	field	Ax:8(1,2)
11	$\int_1^{a} \frac{1}{bu}\, b\, \mathrm{d}u = \int_1^{a} \frac{\mathrm{d}u}{u}$	combine	Eq(10)
12	$\int_1^{a} \frac{\mathrm{d}u}{u} = \ln(a)$	def	Def:14(1)
13	$\ln(ab) - \ln(b) = \ln(a)$	combine	Eq(7, 9, 11, 12)
14	$\ln(a) + \ln(b) = \ln(ab)$	rearrange	Eq(13)
15	$\top$	close	$\to$ I(1, . . . , 14)

Proof of Lemma 5 (ExpProperty)

1	$u,\, v \in \mathbb{R}$	hypothesis	Hyp
2	$a := \exp u,\ \ b := \exp v$	def	def
3	$a,\, b > 0$	def	Def:15(2)
4	$\ln a = u,\ \ \ln b = v$	inv	Def:15(2,3)
5	$ab > 0$	closed	Lem:3(3)
6	$\ln(ab) = \ln a + \ln b = u + v$	Log	Lem:4(3,4,5)
7	$\exp(u+v) = \exp(\ln(ab)) = ab$	inv	Def:15(2,6)
8	$ab = \exp u \cdot \exp v$	def	Eq(2)
9	$\exp u \cdot \exp v = \exp(u+v)$	combine	Eq(7, 8)
10	$\top$	close	$\to$ I(1, . . . , 9)

Citation note (ExpProperty). Row 5 cites **Lem:PosClosedMul(3):** current-table row 3 $(a, b > 0)$ supplies the positivity input to **PosClosedMul**, yielding $ab > 0$ at row 5. Row 6 cites **Lem:LogProperty(3,4,5):** current-table row 3 $(a, b > 0)$ supplies **LogProperty**'s positivity precondition; row 4 $(\ln a = u,\, \ln b = v)$ supplies the logarithmic values; row 5 $(ab > 0)$ makes $\ln(ab)$ well-typed in **LogProperty**'s conclusion. Including row 3 is explicit rather than redundant: **LogProperty**'s hypothesis $a, b : \mathbb{R}_{>0}$ is satisfied by row 3, even though $a =_{\mathrm{def}} \exp u$ already forces $a > 0$.

Proof of Theorem 19 (ImageExpLn)

1	$\forall u,\ u < 0 \lor u \geq 0$	tot	Ax:11(1)
2	$\mathbb{R} = \mathbb{R}_{<0} \cup \mathbb{R}_{\geq 0}$	part	Def:1(1)
3	$\exp(\mathbb{R}) = \exp(\mathbb{R}_{<0}) \cup \exp(\mathbb{R}_{\geq 0})$	ImgUnion	Pos:13(2)
4	$\ln(\exp u) = u$	inv	Def:15(1, 4)
5	$\exp(u) < 1 \land \exp(u) > 0\ (u < 0)$	Sign	Pos:12(4, 3)
6	$\exp(\mathbb{R}_{<0}) =]0,\ 1[$	bij	Pos:11(5)
7	$\exp(u) \geq 1 \land \exp(u) > 0\ (u \geq 0)$	Sign	Pos:12(7, 3)
8	$\exp(\mathbb{R}_{\geq 0}) = [1,\ \infty[$	bij	Pos:11(8)
9	$\exp(\mathbb{R}) =]0,\ \infty[$	combine	Eq(3, 6, 8)
10	$\exp(\mathbb{R}) = \mathbb{R}_{>0}$	def	Def:1(9)
11	$\ln(\mathbb{R}_{>0}) = \mathbb{R}$	bij	Pos:11(1)
12	$\top$	close	$\to$ I(1, ..., 11)

12. Scripts

Each tree lists entries used by a proof, transitively closed, with strengths. `L0` is the root.

12.1. Foundational leaves

Axioms (all ZF).

```
ax:R-add-assoc       (ZF)
ax:R-add-comm        (ZF)
ax:R-add-zero        (ZF)
ax:R-add-inv         (ZF)
ax:R-mul-assoc       (ZF)
ax:R-mul-comm        (ZF)
ax:R-mul-one         (ZF)
ax:R-mul-inv         (ZF)
ax:R-distrib         (ZF)
ax:R-nontriv         (ZF)
ax:R-order           (ZF)
ax:R-add-monotone    (ZF)
ax:R-mul-monotone-pos (ZF)
ax:R-complete        (ZF, Cut)
```

Postulates.

```
Integral:
Pos:IntLinear        (RCA0)
Pos:IntAdd           (RCA0)
Pos:IntMono          (RCA0)
Pos:IntSubst         (RCA0)
Pos:ContInt          (RCA0)
Pos:Similarity       (RCA0)

Differential:
Pos:ProdRule         (RCA0)
Pos:ChainRule        (RCA0)
Pos:InvCont          (RCA0)
Pos:BarrowTangency   (RCA0)

Log/Logic:
Pos:LnBij            (RCA0)
Pos:LnSign           (RCA0)
Pos:ImgUnion         (FOL)

External:
Pos:ContImgBdd       (RCA0)
Pos:ContImgAttain    (WKL0)
Pos:Pi01Sup          (WKL0)
Pos:Sigma01Choice    (RCA0)
Pos:SeqCont          (RCA0)
Pos:HVRolle          (RCA0)
Pos:YokoyamaInv      (RCA0)
```

12.2. Per-result trees

Lemma PosClosedMul (RCA_0)

```
L0 | PosClosedMul                  (RCA0)
L1 |   ax:R-order                  (ZF)
```

```
L1 |   ax:R-mul-monotone-pos          (ZF)
L1 |   ax:R-mul-inv                  (ZF)
```

Lemma **LogProperty** (**RCA$_0$**)

```
L0 | LogProperty                     (RCA0)
L1 |   PosClosedMul                  (RCA0)
L1 |   Pos:IntLinear                 (RCA0)
L1 |   Pos:IntAdd                    (RCA0)
L1 |   Pos:IntSubst                  (RCA0)
L1 |   Def:ln                        (RCA0)
L1 |   ax:R-mul-comm                 (ZF)
L1 |   ax:R-mul-inv                  (ZF)
```

Lemma **ExpProperty** (**RCA$_0$**)

```
L0 | ExpProperty                     (RCA0)
L1 |   LogProperty                   (RCA0)
L1 |   PosClosedMul                  (RCA0)
L1 |   Def:exp                       (RCA0)
```

Theorem **ImageExpLn** (**RCA$_0$**)

```
L0 | ImageExpLn                     (RCA0)
L1 |   ax:R-order                   (ZF)
L1 |   Pos:ImgUnion                 (FOL)
L1 |   Pos:LnBij                    (RCA0)
L1 |   Pos:LnSign                   (RCA0)
L1 |   Def:exp                       (RCA0)
L1 |   Def:ln                        (RCA0)
L1 |   Def:R-pos                     (RCA0)
```

Theorem **MVT**: two proofs, two bounds

Proof A (Barrow, §3; WKL$_0$).

```
L0 | MVT (Proof A)                   (WKL0)
L1 |   EVT                          (WKL0)
L1 |   Pos:BarrowTangency           (RCA0)
L1 |   Def:chord                     (RCA0)
```

Proof B (Rolle-based, H–V; RCA$_0$).

```
L0 | MVT (Proof B)                   (RCA0)
L1 |   Pos:HVRolle                  (RCA0)
L1 |   Def:chord                     (RCA0)
```

The tag RCA$_0$ is the strength of Proof B; Proof A alone is WKL$_0$. Same split for Rolle, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity. Darboux and WeightedMVT are WKL$_0$ throughout.

Theorem **FFTC** (**RCA$_0$**)

```
L0 | FFTC                           (RCA0)
L1 |   BarrowArea                  (RCA0)
L2 |     Pos:IntMono               (RCA0)
L1 |   Pos:IntAdd                    (RCA0)
L1 |   Def:Area                      (RCA0)
L1 |   Def:Ck                        (RCA0)
```

12.3. Transitive closure

```
ImageExpLn                         (RCA0)
|-- ax:R-order                     (ZF)
|-- Pos:ImgUnion                   (FOL)
|-- Pos:LnBij                      (RCA0)
|-- Pos:LnSign                     (RCA0)
```

```
|-- Def:exp                      (RCA0)
|-- Def:ln                       (RCA0)
`-- Def:R-pos                    (RCA0)

ExpProperty                      (RCA0)
|-- LogProperty                  (RCA0)
|-- PosClosedMul                 (RCA0)
`-- Def:exp                      (RCA0)

LogProperty                      (RCA0)
|-- PosClosedMul                 (RCA0)
|-- Pos:IntLinear                (RCA0)
|-- Pos:IntAdd                   (RCA0)
|-- Pos:IntSubst                 (RCA0)
|-- Def:ln                       (RCA0)
|-- ax:R-mul-comm                (ZF)
`-- ax:R-mul-inv                 (ZF)

PosClosedMul                     (RCA0)
|-- ax:R-order                   (ZF)
|-- ax:R-mul-monotone-pos        (ZF)
`-- ax:R-mul-inv                 (ZF)
```

12.4. Strength map

```
STRENGTH  |  ENTRIES
----------+-------------------------------------------------------------
FOL       |  Pos:ImgUnion
----------+-------------------------------------------------------------
RCA0      |  AXIOMS: all fourteen field axioms (ZF, no choice)
|  POSTULATES: IntLinear IntAdd IntMono IntSubst ContInt
|              Similarity ProdRule ChainRule InvCont
|              BarrowTangency LnBij LnSign
|              ContImgBdd Sigma01Choice SeqCont HVRolle YokoyamaInv
|  DEFS: L1 R-pos Ck strong-Ck Diff Area Chord Tangent
|        LocExt Increasing StrictInc StrictDec StrictMono ln exp
|  TOPOLOGY:  (none)
|  DIFF: Fermat Rolle MVT Const Monotone CauchyMVT
|        LHopital Taylor Convexity Inverse
|  INTEG: BarrowArea MVTInt FFTC SecondFTC IBP Subst
|  LOG/EXP: PosClosedMul LogProperty ExpProperty ImageExpLn
----------+-------------------------------------------------------------
WKL0      |  IVT, EVT, Darboux, WeightedMVT
|  (as printed; see EVT tag remark in \S\ref{sec:core-statements})
|  Pos:ContImgAttain, Pos:Pi01Sup
|  IVT (seq), MVT (seq)
|  Hahn-Banach, Weierstrass
----------+-------------------------------------------------------------
ACA0      |  separation (closed convex)
----------+-------------------------------------------------------------
ZF        |  ax:R-add-assoc ... ax:R-complete (all 14)
----------+-------------------------------------------------------------
ZFC       |  Banach-Alaoglu, Tychonoff, well-ordering
----------+-------------------------------------------------------------
```

12.5. Dependency depth

Axioms and postulates are depth 0; definitions depth 0 as primitive notation.

Depth	Entries (selection)
0	14 axioms, 20 postulates, 15 definitions
1	PosClosedMul, Fermat, BarrowArea
2	Rolle, MVT, LogProperty, ImageExpLn, MVTInt, FFTC
3	ExpProperty, Const, Monotone, SecondFTC
4	CauchyMVT, Convexity, IBP, Subst

Longest chain: $ax{:}R\text{-}order \to PosClosedMul \to LogProperty \to ExpProperty$ (length 3). Diff/integ chains reach depth 4.

12.6. Strength propagation

Result	Lower	Upper	Choice
IVT, EVT	WKL_0	WKL_0	none
Fermat	RCA_0	RCA_0	none
Rolle, MVT	RCA_0	RCA_0	none
Const, Monotone	RCA_0	RCA_0	none
CauchyMVT, LHopital, Taylor	RCA_0	RCA_0	none
Convexity	RCA_0	RCA_0	none
Inverse, MVTInt	RCA_0	RCA_0	none
Darboux, WeightedMVT	WKL_0	WKL_0	none
BarrowArea, FFTC	RCA_0	RCA_0	none

Bounds coincide: every RCA_0 tag has an RCA_0 proof in §11 (direct, or Hardin–Velleman, or Yokoyama); every WKL_0 tag has the classical proof. Part I's geometric proofs of Rolle, MVT, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity are at most WKL_0 as written; the gap is disclosed in the EVT remark and does not affect any tag. No proof of Part II uses the Cut axiom.

13. Conclusion

Classical analysis developed in Barrow's geometry (Part I) is classified in reverse mathematics (Part II):

- 14 field axioms: ZF, no choice.
- Postulates: RCA_0-provable or first-order logical.
- RCA_0: Rolle, MVT, Fermat, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity, Inverse, FFTC, SecondFTC, IBP, Subst, ln/exp addition and image.
- WKL_0 as proved here: IVT, EVT, Darboux, WeightedMVT.
- No entry uses Cauchy completeness ($\mathsf{DC}_{\mathbb{R}}$-equiv) or Cut.

13.1. Wider hierarchy

Single-function MVT is RCA_0 [3]. IVT, EVT, and sequential IVT/MVT are WKL_0 [4, 5]. Hahn–Banach and Weierstrass are WKL_0 [4]. Separation is ACA_0 [15]. Banach–Alaoglu, Tychonoff, and well-ordering are ZFC [12, 13, 14]. Every Part I entry is choice-free.

13.2. Dependency analysis

The core chain has depth 3 with longest path $ax{:}R\text{-}order \to PosClosedMul \to LogProperty \to ExpProperty$; differentiation and integration chains reach depth 4.

The load-bearing nodes are ax:R-order, ax:R-mul-inv, Def:ln, Def:exp, Pos:LnBij, and PosClosedMul. The join of predecessor strengths never exceeds WKL_0 and never reaches $\mathsf{ZF} + \mathsf{DC}_{\mathbb{R}}$. Classical theorems are exactly those whose proofs need only area additivity (5), rectangle comparison (27), the differentiating postulates (10)–(11), and (for the four WKL_0 entries) the standard WKL_0-strength principles. No choice principle is needed.

13.3. Formal verification

Lean 4 and Mathlib [10] contain many of these results; about 70.4% of Mathlib declarations depend on Classical.choice, and 13.1% of theorems have a choice-free statement with a choice-dependent proof [11]. The classification above is strictly stronger: it certifies the choice-free proofs.

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